

MATH 127 Calculus for the Sciences

Lecture 27

Today's lecture

Last time

Intervals of Concavity

Inflection point

This time

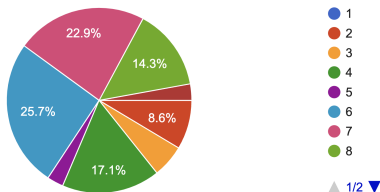
Course note coverage Section 3.4

Curve sketching

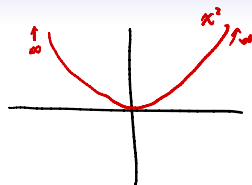
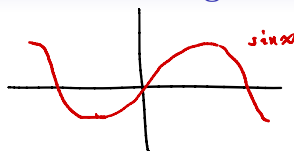
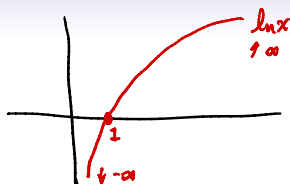
Today's number is...

35 responses

 Copy chart



Curve sketching



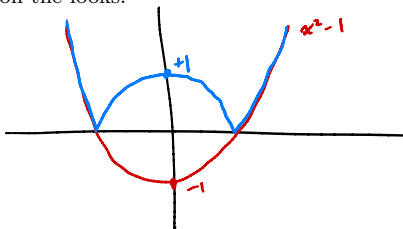
You know how some functions are easy to graph, like

$$\ln(x), \quad \sin(x), \quad x^2$$

But some are more complicated, yet still doable, like

$$2\sin(7x) - 5, \quad \ln(0.5x + 1), \quad \boxed{|x^2 - 1|}.$$

And once you graph them, you can extract information about derivative/concavity based on the looks.



Curve sketching

But sometimes you get crazy functions like

$$5x^{2/3} - 2x^{5/3}.$$

If you want to graph this, you can try to work backwards...

1. Find information about derivative/concavity/critical point/etc using algebra.
2. Then draw the graph based on these information.

That's it... the rest is just practice.

Curve sketching

Example Sketch the curve

$$f(x) = x \ln(x)$$

1. The domain is: $(0, \infty)$
2. A few obvious points on this curve are: $f(1) = 1 \cdot \ln(1) = 0$. $f(e) = e \ln e = e$
3. The derivative is: $\ln x + 1$.
4. The critical points are: $\ln x + 1 = 0$ then $\ln x = -1$, so $x = e^{-1}$ is a critical point.
5. The second derivative is: $\frac{1}{x}$
6. The inflection points are: no inflection points.

Curve sketching

Example Sketch the curve

$$f(x) = x \ln(x)$$

1. The domain is: $(0, \infty)$
2. A few obvious points on this curve are: $(1, 0)$ and a limit point of $(0, 0)$
3. The derivative is: $\ln(x) + 1$
4. The critical points are: $x = 1/e$; on the left $f'(x)$ is negative; on the right $f'(x)$ is positive.
5. The second derivative is: $\frac{1}{x}$; which is positive on the domain $(0, \infty)$

limit point of $(0, 0)$:

This means

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{x}{1/\ln x} \quad \text{or} \quad \boxed{\frac{\ln x}{1/x}} \quad \text{easier}$$

$0^+(-\infty)$ bad

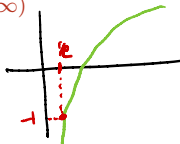
$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} -x = 0$$

when $0 < x < 1/e$,
 $\ln x < -1$

$$f'(x) = \ln x + 1 < 0$$

when $x > 1/e$
then $\ln x > -1$

$$\text{so } f'(x) = \ln x + 1 > 0$$



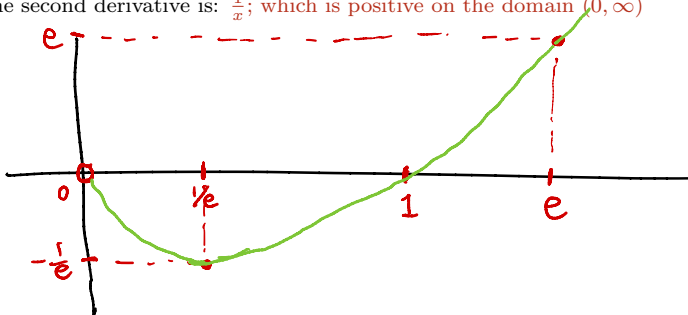
Curve sketching

Example Sketch the curve

$$f(x) = x \ln(x)$$

$$\begin{aligned} f\left(\frac{1}{e}\right) &= \frac{1}{e} \cdot \ln \frac{1}{e} \\ &= \frac{1}{e} \cdot (-1) = -\frac{1}{e} \end{aligned}$$

1. The domain is: $(0, \infty)$
2. A few obvious points on this curve are: $(1, 0)$ and a limit point of $(0, 0)$
3. The derivative is: $\ln(x) + 1$
4. The critical points are: $x = 1/e$; on the left $f'(x)$ is negative; on the right $f'(x)$ is positive.
5. The second derivative is: $\frac{1}{x}$; which is positive on the domain $(0, \infty)$



Curve sketching

Example Sketch the curve

$$f(x) = \frac{x^3}{x^2 + 1}, \text{ you are told that } f'(x) = \frac{x^2(x^2 + 3)}{(x^2 + 1)^2} \text{ and } f''(x) = \frac{2x(3 - x^2)}{(x^2 + 1)^3}$$

1. The domain is:
2. A few obvious points on this curve are:
3. The critical points are:
4. The inflection points are:

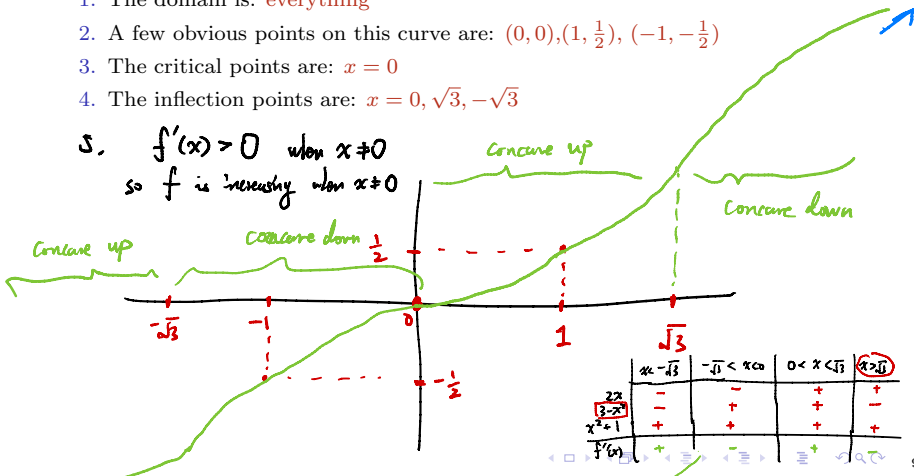
Curve sketching

Example Sketch the curve

$$f(x) = \frac{x^3}{x^2 + 1}, \text{ you are told that } f'(x) = \frac{x^2(x^2 + 3)}{(x^2 + 1)^2} \text{ and } f''(x) = \frac{2x(3 - x^2)}{(x^2 + 1)^3}$$

1. The domain is: **everything**
2. A few obvious points on this curve are: $(0, 0), (1, \frac{1}{2}), (-1, -\frac{1}{2})$
3. The critical points are: $x = 0$
4. The inflection points are: $x = 0, \sqrt{3}, -\sqrt{3}$

5. $f'(x) > 0$ when $x \neq 0$
so f is increasing when $x \neq 0$



Curve sketching

Example Sketch the curve

$$f(x) = xe^{-1/x}$$

For Wednesday

We'll do more curve sketching. I want you to come up with difficult functions you want me to sketch. And I'll do it on the spot on Wednesday to show that I'm not a fake mathematician.